SANJAY BAIRWA

Full Stack - Flutter Developer — Mobile Application Developer

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Professional Summary

Flutter Developer specializing in cross-platform mobile application development using Dart, Flutter, Firebase, REST APIs, and modern state management. Experienced in developing responsive mobile interfaces, integrating authentication and real-time databases, consuming RESTful APIs, and implementing Clean Architecture. Strong foundation in Data Structures and Algorithms, Object-Oriented Programming, and scalable application development.

Technical Skills

Languages: Dart, JavaScript, TypeScript, Python, HTML5, CSS3
Mobile Development: Flutter, Android Development, Cross-Platform Development, Responsive UI, Provider, Riverpod
Backend & APIs: Node.js, Express.js, FastAPI, REST APIs, RESTful Web Services, JSON
Firebase: Firebase Authentication, Cloud Firestore, Firebase Storage, Firebase Cloud Messaging (FCM)
Databases: MongoDB, PostgreSQL, MySQL, SQLite, Hive
AI & Computer Vision: TensorFlow, OpenCV, MediaPipe, Computer Vision, Generative AI, Gemini API
Architecture & Tools: Clean Architecture, Git, GitHub, Docker, Android Studio, VS Code, Postman

Projects

SignSpeak – Sign Language Communication Platform 2025 – Present
Flutter • Firebase • Riverpod • Python • FastAPI • MediaPipe • TensorFlow

- Developed a cross-platform Flutter application for sign-language communication with real-time messaging, video calling, and AI-assisted sign recognition.
- Integrated Firebase Authentication, Cloud Firestore, Firebase Storage, and FCM for authentication, real-time data synchronization, media storage, and push notifications.
- Built a Python computer-vision pipeline using MediaPipe and OpenCV for hand-landmark extraction and TensorFlow-based sign recognition.
- Integrated Flutter with FastAPI backend services through REST APIs and implemented Riverpod for application state management.

Video Forensics Analyzer 2026
Python • OpenCV • FFmpeg • FFprobe

- Developed a video analysis pipeline to detect freeze frames, duplicate frames, missing frames, corrupted frames, scene changes, and potential frame-level modifications.
- Implemented frame-level processing with Python and OpenCV, using FFmpeg and FFprobe for video decoding, metadata extraction, and media processing.
- Designed an evidence-fusion module that combines detector results and generates structured JSON reports with anomaly types, frame ranges, timestamps, and detection scores.

TaskFlow – AI-Powered Task Management Application 2026 [APK Demo]
React Native • Node.js • Express.js • MongoDB • REST API • AI

- Developed a cross-platform task management application with task creation, editing, deletion, status tracking, filtering, and persistent backend synchronization.
- Built and integrated a Node.js and Express.js REST API with MongoDB for task and user data management.
- Integrated AI-powered functionality to assist users with task organization and productivity workflows through intelligent task processing.
- Implemented API-driven data synchronization between the mobile application and backend services and deployed the backend using Render.

Education

B.Tech in Computer Science 2023 – 2027
Guru Ghasidas University, Bilaspur — CGPA: 7.5/10 (Expected)
Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Object-Oriented Programming

Certifications

Salesforce Administrator & App Builder Sep 2026 [Certificate]
Introduction to Generative AI Studio Sep 2026 [Certificate]

Achievements

- Developed multiple mobile applications integrating REST APIs, Firebase, authentication, cloud databases, and modern state management.
- Built AI and computer-vision solutions using Python, MediaPipe, OpenCV, TensorFlow, and Generative AI technologies.